Drixit's GRE Notes

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Information Station

Information provided by GRE practice materials

Rest mass of the electron	$m_e =$	9.11×10^{-31}	kg	10^{-15}	femto	f
Magnitude of the electron charge	$e =$	1.60×10^{-19}	C	10^{-12}	pico	p
Avagadro's number	$N_A =$	6.022×10^{23}	mol^{-1}	10^{-9}	nano	n
Universal gas constant	$R =$	8.31	$\mathrm{J\,mol^{-1}\,K^{-1}}$	10^{-6}	micro	µ
Boltzmann's constant	$k =$	1.38×10^{-23}	J/k	10^{-3}	milli	m
Speed of light	$c =$	3.00×10^{8}	m/s	10^{-2}	centi	c
Planck's constant	$h =$	6.63×10^{-34}	J s	10^{-1}	deci	d
	$=$	4.14×10^{-15}	eV s	10^{3}	kilo	k
	$\hbar =$	$\frac{h}{2\pi}$		10^{6}	mega	M
	hc $=$	1240	eV nm	10^{9}	giga	G
Vacuum permittivity	$\varepsilon_0 =$	8.85×10^{-12}	$\mathrm{C^2\ /\ N\,m^2}$	10^{12}	tera	T
Vacuum permeability	$\mu_0 =$	$4\pi \times 10^{-7}$	$\mathrm{Tm\,A^{-1}}$	10^{15}	peta	P
Universal gravitational constant	$G =$	6.67×10^{-11}	$\mathrm{N\,m^2\,kg^{-2}}$			
Acceleration due to gravity	$g =$	9.80	$\mathrm{m\,s^{-2}}$			
Standard temperature and pressure	STP:	0°C and 1 atm				
	1 atm $=$	1.0×10^{5}	$\mathrm{N\,m^{-2}}$			
			Pa			
1 angstrom	Å $=$	1×10^{-10}	m			
	Å $=$	0.1	nm			

Table 1: Constants and prefixes for powers of 10

Rod of mass M and length L	$I = \frac{1}{12}ML^2$
Disk of mass M and radius R	$I = \frac{1}{2}MR^2$
Solid sphere of mass M and radius R	$I = \frac{2}{5}MR^2$

Table 2: Rotational inertia about center of mass for common shapes

Practice Questions

1. Two objects sliding on a frictionless surface, as represented below, collide and stick together. How much kinetic energy is converted to heat during the collision? (Educational Testing Service, 2024, Question 1)

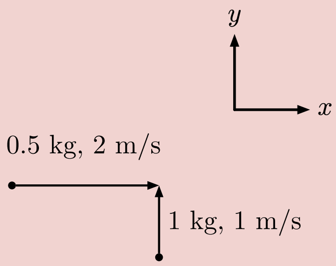


Figure 1: Collision of two objects on a frictionless surface

Expected knowledge and equations:

"In an isolated system ... momentum is always conserved, whether the collisoin is elastic or not."

Energy	**Momentum**	**Collision Classification**
Kinetic Energy (KE) for mass m moving at speed v	Momentum ($\vec{\boldsymbol{P}}$) for mass m moving at velocity $\vec{\boldsymbol{v}}$	Elastic, Inelastic, Perfectly Inelastic
$\mathrm{KE} = \frac{1}{2}mv^2$ $\mathrm{KE} = \frac{1}{2}m\dot{\vec{\boldsymbol{v}}}$	$\vec{\boldsymbol{P}} = m\vec{\boldsymbol{v}}$	

Table 3: Energy, momentum, and collision classifications

The KE lost in an Inelastic/Perfectly Inelastic collision is converted to other forms of energy, such as heat and sound.

Collision	**Equations**, where 1: before collision, 2: after collision
Elastic	KE is conserved $\mathrm{KE}_1 = \mathrm{KE}_2$ Momentum is conserved $\vec{\boldsymbol{P}}_1 = \vec{\boldsymbol{P}}_2$
Inelastic	KE is not conserved $\mathrm{KE}_1 > \mathrm{KE}_2$ Momentum is conserved $\vec{\boldsymbol{P}}_1 = \vec{\boldsymbol{P}}_2$ Converted Energy ($\mathrm{E_C}$) = KE_1 - KE_2
Perfectly Inelastic	KE is not conserved $\mathrm{KE}_1 > \mathrm{KE}_2$ Momentum is conserved and objects stick together Converted Energy ($\mathrm{E_C}$) = KE_1 - KE_2

Table 4: Collision classifications and their corresponding equations

(Young et al., 2016, Chapter 8.2, p.248-250)

Given:

Mass (kg), Velocity (m/s), let x be the $\boldsymbol{\hat{\imath}}$ direction and y be the $\boldsymbol{\hat{\jmath}}$ direction

(Mass of object α, velocity of object α)= $(m_\alpha, \overrightarrow{\boldsymbol{v_\alpha}}) = (0.5, 2\boldsymbol{\hat{\imath}} + 0\boldsymbol{\hat{\jmath}}) = \left(\frac{1}{2}, 2\boldsymbol{\hat{\imath}} + 0\boldsymbol{\hat{\jmath}}\right)$

(Mass of object β, velocity of object β)= $\left(m_\beta, \overrightarrow{\boldsymbol{v_\beta}}\right) = (1, 0\boldsymbol{\hat{\imath}} + 1\boldsymbol{\hat{\jmath}})$

Worked Solution:
Assumptions: "stick" ⇒ Perfectly Inelasitic collision ⇒ Kinetic Energy (KE) is not conserved i.e. $\mathrm{E_C}$ exists and objects stick together i.e. the combined mass after the collision is $m_\alpha + m_\beta$.

$$m_\alpha + m_\beta = \frac{1}{2} + 1 = \frac{3}{2} \tag{i}$$

The KE converted to heat ($\mathrm{E_C}$) during the collision of two objects on a frictionless surface is as follows,

$$\begin{aligned} \mathrm{E_C} &= \mathrm{KE_1} - \mathrm{KE_2} \\ &= \left(\left(\frac{1}{2}\right)m_\alpha \dot{\overrightarrow{\boldsymbol{v_\alpha}}} + \left(\frac{1}{2}\right)m_\beta \dot{\overrightarrow{\boldsymbol{v_\beta}}}\right) - \left(\left(\frac{1}{2}\right)(m_\alpha + m_\beta)\dot{\overrightarrow{\boldsymbol{v_f}}}\right) \end{aligned} \tag{ii}$$

where $\overrightarrow{\boldsymbol{v_f}}$ is the final velocity of the combined mass after the collision and,

$$\begin{aligned} \dot{\overrightarrow{\boldsymbol{v_\alpha}}} &= \overrightarrow{\boldsymbol{v_\alpha}} \cdot \overrightarrow{\boldsymbol{v_\alpha}} \\ &= < 2\boldsymbol{\hat{\imath}} + 0\boldsymbol{\hat{\jmath}} > \cdot < 2\boldsymbol{\hat{\imath}} + 0\boldsymbol{\hat{\jmath}} > \\ &= 2 \cdot 2 + 0 \cdot 0 \\ &= 4 \end{aligned} \tag{iii}$$

similarly,

$$\begin{aligned} \dot{\overrightarrow{\boldsymbol{v_\beta}}} &= \overrightarrow{\boldsymbol{v_\beta}} \cdot \overrightarrow{\boldsymbol{v_\beta}} \\ &= < 0\boldsymbol{\hat{\imath}} + 1\boldsymbol{\hat{\jmath}} > \cdot < 0\boldsymbol{\hat{\imath}} + 1\boldsymbol{\hat{\jmath}} > \\ &= 0 \cdot 0 + 1 \cdot 1 \\ &= 1 \end{aligned} \tag{iv}$$

Using the conservation of momentum we can find $\overrightarrow{\boldsymbol{v_f}}$

$$\begin{aligned} \vec{\boldsymbol{P}}_1 &= \vec{\boldsymbol{P}}_2 \\ m_\alpha \overrightarrow{\boldsymbol{v_\alpha}} + m_\beta \overrightarrow{\boldsymbol{v_\beta}} &= (m_\alpha + m_\beta)\overrightarrow{\boldsymbol{v_f}} \\ \frac{m_\alpha \overrightarrow{\boldsymbol{v_\alpha}} + m_\beta \overrightarrow{\boldsymbol{v_\beta}}}{(m_\alpha + m_\beta)} &= \overrightarrow{\boldsymbol{v_f}} \text{ , where } m_\alpha + m_\beta \neq 0 \\ \frac{\frac{1}{2} \cdot < 2\boldsymbol{\hat{\imath}} + 0\boldsymbol{\hat{\jmath}} > + 1 \cdot < 0\boldsymbol{\hat{\imath}} + 1\boldsymbol{\hat{\jmath}} >}{\frac{1}{2} + 1} &= \overrightarrow{\boldsymbol{v_f}} \\ \frac{< \boldsymbol{\hat{\imath}} + 0\boldsymbol{\hat{\jmath}} > + < 0\boldsymbol{\hat{\imath}} + 1\boldsymbol{\hat{\jmath}} >}{\frac{3}{2}} &= \overrightarrow{\boldsymbol{v_f}} \\ \frac{2}{3} < \boldsymbol{\hat{\imath}} + \boldsymbol{\hat{\jmath}} > &= \overrightarrow{\boldsymbol{v_f}} \end{aligned} \tag{v}$$

it follows that,

$$\begin{aligned} \dot{\overrightarrow{\boldsymbol{v_f}}} &= \overrightarrow{\boldsymbol{v_f}} \cdot \overrightarrow{\boldsymbol{v_f}} \\ &= < \frac{2}{3}\boldsymbol{\hat{\imath}} + \frac{2}{3}\boldsymbol{\hat{\jmath}} > \cdot < \frac{2}{3}\boldsymbol{\hat{\imath}} + \frac{2}{3}\boldsymbol{\hat{\jmath}} > \\ &= \frac{2}{3} \cdot \frac{2}{3} + \frac{2}{3} \cdot \frac{2}{3} \\ &= \frac{8}{9} \end{aligned} \tag{vi}$$

therefore,

$$\begin{aligned}
\mathrm{E_C} &= \left(\left(\frac{1}{2}\right)m_\alpha\dot{\vec{\boldsymbol{v_\alpha}}} + \left(\frac{1}{2}\right)m_\beta\dot{\vec{\boldsymbol{v_\beta}}}\right) - \left(\left(\frac{1}{2}\right)(m_\alpha + m_\beta)\dot{\vec{\boldsymbol{v_f}}}\right) \\
&= \left(\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)(4) + \left(\frac{1}{2}\right)(1)(1)\right) - \left(\left(\frac{1}{2}\right)\left(\frac{3}{2}\right)\left(\frac{8}{9}\right)\right) \\
&= \left(1 + \frac{1}{2}\right) - \frac{3}{4}\cdot\frac{8}{9} \\
&= \frac{3}{2} - \frac{2}{3} \qquad \text{(vii)} \\
&= \frac{3}{2}\cdot\frac{3}{3} - \frac{2}{3}\cdot\frac{2}{2} \\
&= \frac{9}{6} - \frac{4}{6} \\
&= \frac{5}{6}J
\end{aligned}$$

■

2. Two simple pendulums A and B consist of identical masses suspended from strings of length L_A and L_B, respectively. The two pendulums oscillate in equal gravitational fields. If the period of pendulum B is twice the period of pendulum A, what is the ratio of the lengths of the two pendulums? (Educational Testing Service, 2024, Question 2) No illustration is provided.

Expected knowledge and equations:

Simple Harmonics and Simple Pendulum Equations

Given:

Period of pendulum A: T_A , Length of pendulum A: L_A

Period of pendulum B: $T_B = 2 \cdot T_A$, Length of pendulum B: L_B

Equal gravitational fields: $g_A = g_B = g$

Worked Solution:

■

Block Examples

simple block

question block

Expected knowledge and equations:
knowledge block

Given:
given block

Worked Solution:
solution block ■

Supplemental Math Content

Vector Transformation $\mathbb{R}^2$

Rotation of a vector in 2D

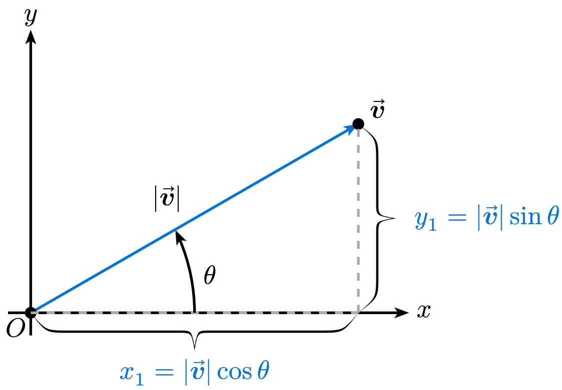


Figure 2: Vector $\vec{\boldsymbol{v}}$

For a vector $\vec{\boldsymbol{v}}(x, y)$ with coordinates,

$$\vec{\boldsymbol{v}}(x, y) = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} |\vec{\boldsymbol{v}}| \cos\theta \\ |\vec{\boldsymbol{v}}| \sin\theta \end{pmatrix} \tag{1}$$

where

$$|\vec{\boldsymbol{v}}| = \sqrt{(x_1)^2 + (y_1)^2} \tag{2}$$

is the magnitude of $\vec{\boldsymbol{v}}$ and θ is the angle between the vector and the positive x-axis. We can rotate $\vec{\boldsymbol{v}}$ by an angle φ using the angle sum rule.

$$\begin{aligned} \cos(\alpha + \beta) &= \cos\alpha\cos\beta - \sin\alpha\sin\beta \\ \sin(\alpha + \beta) &= \cos\alpha\sin\beta + \sin\alpha\cos\beta \end{aligned} \tag{3}$$

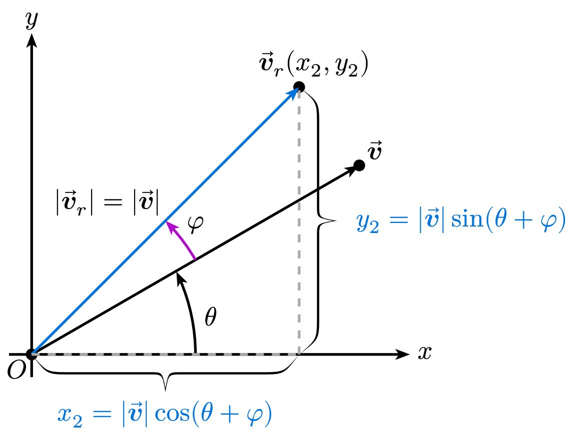


Figure 3: Vector $\vec{\boldsymbol{v}}$ after a rotation by angle φ

Applying the angle sum rule to the components of rotated vector $\vec{\boldsymbol{v}}_r$, we get:

$$
\begin{aligned}
\vec{\boldsymbol{v}}_r = \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} &= \begin{pmatrix} |\vec{\boldsymbol{v}}| \cos(\theta + \varphi) \\ |\vec{\boldsymbol{v}}| \sin(\theta + \varphi) \end{pmatrix} \\
&= \begin{pmatrix} |\vec{\boldsymbol{v}}| \ (\cos\theta \cos\varphi - \sin\theta \sin\varphi) \\ |\vec{\boldsymbol{v}}| \ (\cos\theta \sin\varphi + \sin\theta \cos\varphi) \end{pmatrix} \\
&= \begin{pmatrix} |\vec{\boldsymbol{v}}| \cos\theta \cos\varphi - |\vec{\boldsymbol{v}}| \sin\theta \sin\varphi \\ |\vec{\boldsymbol{v}}| \cos\theta \sin\varphi + |\vec{\boldsymbol{v}}| \sin\theta \cos\varphi \end{pmatrix} \\
&= \begin{pmatrix} x_1 \cos\varphi - y_1 \sin\varphi \\ x_1 \sin\varphi + y_1 \cos\varphi \end{pmatrix}
\end{aligned} \tag{4}
$$

Similarly, we can write $\vec{\boldsymbol{v}}_r$ in matrix form:

$$
\begin{aligned}
\vec{\boldsymbol{v}}_r &= \begin{pmatrix} x_1 \cos\varphi - y_1 \sin\varphi \\ x_1 \sin\varphi + y_1 \cos\varphi \end{pmatrix} \\
&= \begin{pmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{pmatrix} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} \\
&= \begin{pmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{pmatrix} \vec{\boldsymbol{v}} \\
&= A\vec{\boldsymbol{v}}
\end{aligned} \tag{5}
$$

Where A is our rotation matrix for a counterclockwise rotation by angle φ. Note, for a clockwise rotation, we would use $\varphi_- = -\varphi$ in the rotation matrix, which would give us:

$$
\begin{aligned}
A' &= \begin{pmatrix} \cos\varphi_- & -\sin\varphi_- \\ \sin\varphi_- & \cos\varphi_- \end{pmatrix} \\
&= \begin{pmatrix} \cos(-\varphi) & -\sin(-\varphi) \\ \sin(-\varphi) & \cos(-\varphi) \end{pmatrix} \\
&= \begin{pmatrix} \cos\varphi & \sin\varphi \\ -\sin\varphi & \cos\varphi \end{pmatrix}
\end{aligned} \tag{6}
$$

For a visual example see: **Drixit's Desmos Graph of Vector Rotation in 2D**

Parametric Equations

Lines

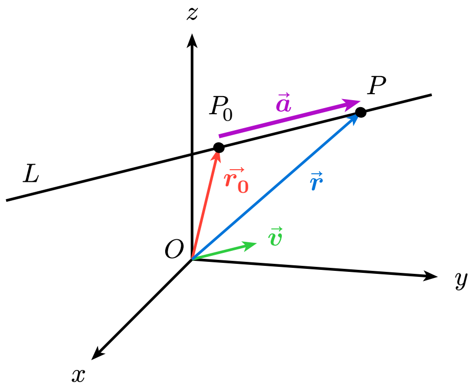


Figure 4: Parametic Line in $\mathbb{R}^3$

Let $P(x, y, z)$ be an arbitrary point in $\mathbb{R}^3$ on line L with a parallel unit vector $\vec{\boldsymbol{v}}$ (does not need a unit vector), and let $\vec{\boldsymbol{r}}_0$ ad $\vec{\boldsymbol{r}}$ be the position vectors of P_0 and P respectively. If $\vec{\boldsymbol{a}}$ is the vector representation $\overrightarrow{\boldsymbol{P_0P}}$, as shown in **Figure 4**, then the Triangle Law for vector addition gives us,

$$\vec{\boldsymbol{a}} = \vec{\boldsymbol{r}} - \vec{\boldsymbol{r}}_0 \Leftrightarrow \vec{\boldsymbol{r}} = \vec{\boldsymbol{r}}_0 + \vec{\boldsymbol{a}} \tag{7}$$

But, since $\vec{\boldsymbol{a}}$ and $\vec{\boldsymbol{v}}$ are parallel vectors, there is a scalar t such that,

$$\vec{\boldsymbol{a}} = t\vec{\boldsymbol{v}} \tag{8}$$

By substituting **Equation 8** into **Equation 7**, we get the vector equation of a line in $\mathbb{R}^3$ as,

$$\vec{\boldsymbol{r}} = \vec{\boldsymbol{r}}_0 + t\vec{\boldsymbol{v}} \tag{9}$$

The parametic form of a line through the point $P_0(x_0, y_0, z_0)$ with a parallel vector $\vec{\boldsymbol{v}} = \langle v_x, v_y, v_z \rangle$ is given by:

$$\vec{\boldsymbol{r}}(t) = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x_0 \\ y_0 \\ z_0 \end{pmatrix} + t \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} \tag{10}$$

A **line segment** from $\vec{\boldsymbol{r}}_0$ to some new $\vec{\boldsymbol{r}}_1$ is given by,

$$\vec{\boldsymbol{r}}(t) = (1-t)\vec{\boldsymbol{r}}_0 + t\vec{\boldsymbol{r}}_1 \text{ for } 0 \leq t \leq 1 \tag{11}$$

(Stewart, 2016, Chapter 12.5)

Proof of Equation 11:
Using **Equation 9**,

$$\vec{\boldsymbol{r}} = \vec{\boldsymbol{r}}_0 + t\vec{\boldsymbol{v}} \tag{i}$$

Let $\vec{\boldsymbol{v}} = \vec{\boldsymbol{r}}_1 - \vec{\boldsymbol{r}}_0$ (see **Figure 4**). Then, we obtain a line segment equation from $\vec{\boldsymbol{r}}_0$ to $\vec{\boldsymbol{r}}_1$

$$\begin{aligned} \vec{\boldsymbol{r}}(t) &= \vec{\boldsymbol{r}}_0 + t(\vec{\boldsymbol{r}}_1 - \vec{\boldsymbol{r}}_0) \\ &= \vec{\boldsymbol{r}}_0 + t\vec{\boldsymbol{r}}_1 - t\vec{\boldsymbol{r}}_0 \\ &= \vec{\boldsymbol{r}}_0 - t\vec{\boldsymbol{r}}_0 + t\vec{\boldsymbol{r}}_1 \\ &= (1-t)\vec{\boldsymbol{r}}_0 + t\vec{\boldsymbol{r}}_1 \text{ for } 0 \leq t \leq 1 \end{aligned} \tag{ii}$$

For a visual example see: **Drixit's Desmos Graph of a Line Segment** ■

Worked Example of Parametric Lines:
The following was used to create **Figure 4**. Note ($'$) is not used as a derivative here.

For a line L with starting point $P_0'(-1,1,1)$ and end point $P_1'(4,4,4)$ we'll calculate the parallel vector $\vec{\boldsymbol{v}}'$.

$$\begin{aligned} \vec{\boldsymbol{v}}' &= \langle x_1 - x_0, y_1 - y_0, z_1 - z_0 \rangle \\ &= \langle 4-(-1), 4-1, 4-1 \rangle \\ &= \langle 5,3,3 \rangle \end{aligned} \tag{i}$$

As the unit vector $\vec{\boldsymbol{v}}$:

$$\begin{aligned} \vec{\boldsymbol{v}} &= \frac{\vec{\boldsymbol{v}}'}{|\vec{\boldsymbol{v}}'|} \\ &= \frac{\langle 5,3,3 \rangle}{\sqrt{5^2+3^2+3^2}} \end{aligned} \tag{ii}$$

Using **Equation 9** we obtain our parametric line L,

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} + t' \begin{pmatrix} 5 \\ 3 \\ 3 \end{pmatrix} \tag{iii}$$

For $0 < t' < 1$ where $t_0' = 0$ gives P_0' and $t_1' = 1$ gives P_1'.

Next we want to find 2 points on L which are asthetic and provide enough space for labels. We can choose $t_2' = \frac{1}{2}$ and $t_3' = \frac{5}{6}$. These values were choosen with guess and check.

For $t_2' = \frac{1}{2}$, we obtain **Figure 4**'s P_0 as,

$$\begin{aligned} P_0 \equiv \begin{pmatrix} x \\ y \\ z \end{pmatrix} &= \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} + \frac{1}{2} \cdot \begin{pmatrix} 5 \\ 3 \\ 3 \end{pmatrix} \\ &= \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} + \begin{pmatrix} \frac{5}{2} \\ \frac{3}{2} \\ \frac{3}{2} \end{pmatrix} \\ &= \begin{pmatrix} \frac{3}{2} \\ \frac{5}{2} \\ \frac{5}{2} \end{pmatrix} \\ &= \vec{\boldsymbol{r}}_0 \end{aligned} \tag{iv}$$

For $t_3' = \frac{5}{6}$, we obtain **Figure 4**'s P as,

$$P \equiv \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} + \frac{5}{6} \cdot \begin{pmatrix} 5 \\ 3 \\ 3 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} + \begin{pmatrix} \frac{25}{6} \\ \frac{15}{6} \\ \frac{15}{6} \end{pmatrix} = \begin{pmatrix} \frac{19}{6} \\ \frac{21}{6} \\ \frac{21}{6} \end{pmatrix} = \vec{\boldsymbol{r}} \tag{v}$$

Lastly we can obtain the vector $\vec{\boldsymbol{a}}$ as,

$$\vec{\boldsymbol{a}} = \vec{\boldsymbol{r}} - \vec{\boldsymbol{r}}_0 = \begin{pmatrix} \frac{19}{6} \\ \frac{21}{6} \\ \frac{21}{6} \end{pmatrix} - \begin{pmatrix} \frac{3}{2} \\ \frac{5}{2} \\ \frac{5}{2} \end{pmatrix} = \begin{pmatrix} \frac{5}{3} \\ 1 \\ 1 \end{pmatrix} \tag{vi}$$

Though to plot with Typst we need to shift the z-values as follows,

$$P_0\mathrm{z}+ = \begin{pmatrix} \frac{3}{2} \\ \frac{5}{2} \\ \frac{5}{2} + 0.1 \end{pmatrix} \qquad P\mathrm{z}+ = \begin{pmatrix} \frac{19}{6} \\ \frac{21}{6} \\ \frac{21}{6} + 0.1 \end{pmatrix} \tag{vii}$$

■

Derivatives of Parametric Lines

Derivative

$$\frac{d\vec{r}}{dt} = \vec{r}'(t) = \lim_{h \to 0} \frac{\vec{r}(t+h) - \vec{r}(t)}{h} \tag{12}$$

Theorem: If $\vec{r}(t) = \langle f(t), g(t), h(t) \rangle = f(t)i + g(t)j + h(t)k$, where $f, g,$ and h are differentiable functions, then

$$\vec{r}'(t) = \langle f'(t), g'(t), h'(t) \rangle = f'(t)i + g'(t)j + h'(t)k \tag{13}$$

(Stewart, 2016, Chapter 13.2)

Worked Example: Derivative of a Parametric Circle

The equations of a circle of radius r centered at the origin in $\mathbb{R}^2$ are given by:

$$\begin{aligned} r^2 &= x^2 + y^2 \\ x &= r\cos(\theta) \\ y &= r\sin(\theta) \end{aligned} \tag{i}$$

Our parametic representation of the circle is then,

$$\vec{r}(\theta) = \langle r\cos(\theta), r\sin(\theta) \rangle \tag{ii}$$

Taking the derivative with respect to θ using **Equation 13**, we get,

$$\vec{r}'(\theta) = \langle -r\sin(\theta), r\cos(\theta) \rangle \tag{iii}$$

If we were to take this further and apply what we know about parametric lines we can construct a tangent line to the circle at a point P_0.

Using **Equation 9** we let $\vec{r}_0 = \vec{r}(\theta)$ and $\vec{v} = \vec{r}'(\theta)$, now our equation of the tangent line is,

$$\begin{aligned} \vec{r}_{\text{tangent}} &= \vec{r}_0 + t\vec{v} \\ &= \vec{r} + t\vec{r}' \end{aligned} \tag{iv}$$

For a visual example see: **Drixit's Desmos Graph of Tangent Line to a Parametric Circle**

■

Supplemental Physics Content

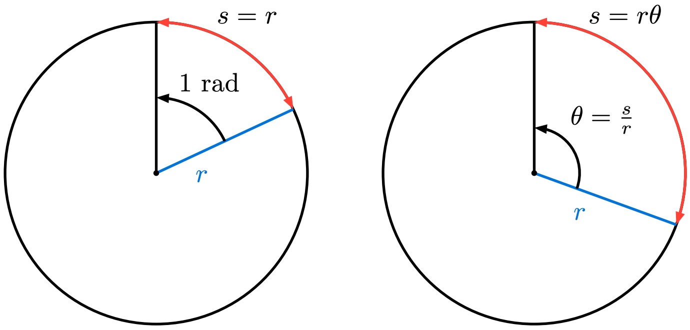
Uniform Circular Motion

Uniform Circular Motion
- When a particle moves in a circle with constant speed

Rotation of Rigid Bodies

Radian
- **1 radian** is the angle at which the **arc length** $\boldsymbol{s}$ has the same length as the **radius** $\boldsymbol{r}$
- An **angle** $\boldsymbol{\theta}$ in radians is the ratio of the **arc length** $\boldsymbol{s}$ to the **radius** $\boldsymbol{r}$

$$s = r\theta \tag{14}$$



(a) Radian (b) Angle θ

Figure 5: Measuring angles in radians

"Always use radians when relating linear and angular quantiites i.e. $s = r\theta$"
(Young et al., 2016, Chapter 9.3, p.281)

Degrees

$$1 \text{ radian} = \frac{360°}{2\pi} = \frac{180°}{\pi} = 57.2958° \tag{15}$$

Revolutions

$$1 \text{ radian} = \frac{1 \text{ revolution}}{2\pi} \Leftrightarrow 1 \text{ revolution} = 2\pi \text{ radian} \tag{16}$$

Similarly

$$1\frac{\text{rev}}{\text{min}} = 1\text{rpm} = \frac{2\pi}{60}\frac{\text{rad}}{s} = \frac{2\pi \text{ rad}}{\text{min}} \tag{17}$$

Average Angular Velocity $\vec{\boldsymbol{\omega}}_{\text{av}}$
- where av-z represents **"av" average**, **"z"** for the z-component of Angular Velocity $\vec{\boldsymbol{\omega}}$ of a rigid body rotating **about the z-axis**, and $\boldsymbol{t}$ for **time**

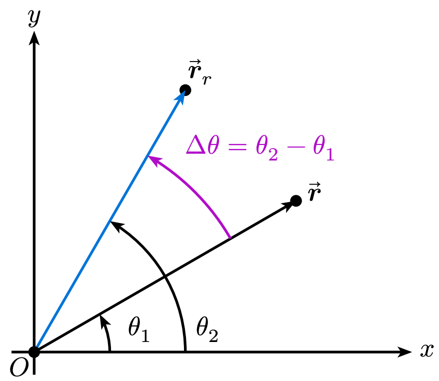


Figure 6: Average angular velocity of a rigid body with position vector $\vec{\boldsymbol{r}}$ at time t_1 and its rotated position vector $\vec{\boldsymbol{r}}_r$ at time t_2

$$\omega_{\text{av-z}} = \frac{\theta_2 - \theta_1}{t_2 - t_1} = \frac{\Delta\theta}{\Delta t} \tag{18}$$

Instantaneous Angular Velocity $\vec{\boldsymbol{\omega}}$

$$\omega_z = \lim_{\Delta t \to 0} \frac{\Delta\theta}{\Delta t} = \frac{d\theta}{dt} \tag{19}$$

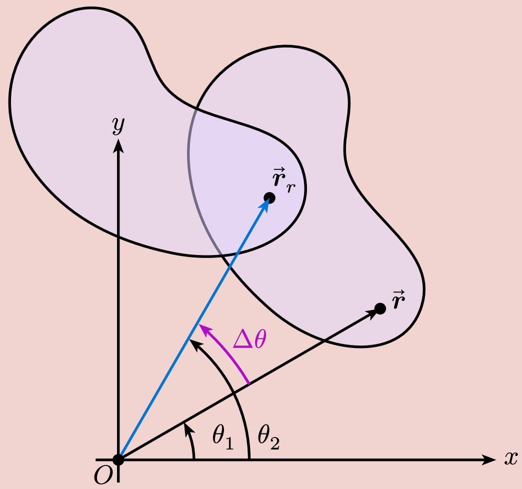


Figure 7: Angular velocity ω as an attribute of the entire rigid body

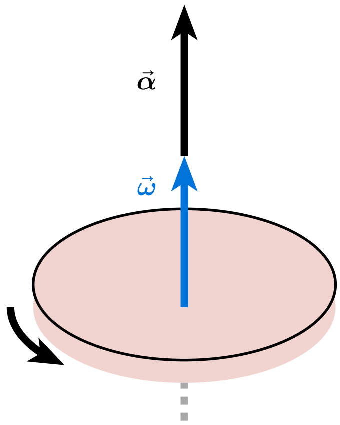
"Angular velocity is an attibute of the entire rigid body, not any one part"
(Young et al., 2016, Chapter 9.1, p.276)

Average Angular Acceleration $\vec{\boldsymbol{\alpha}}_{\text{av}}$

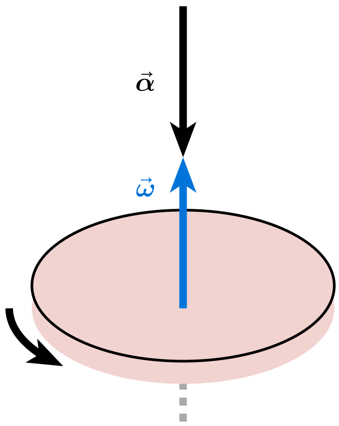
$$\alpha_{\text{av-z}} = \frac{\omega_{2z} - \omega_{1z}}{t_2 - t_1} = \frac{\Delta\omega_z}{\Delta t} \tag{20}$$

Instantaneous Angular Acceleration $\vec{\boldsymbol{\alpha}}$

$$\alpha_z = \lim_{\Delta t \to 0} \frac{\Delta\omega_z}{\Delta t} = \frac{d\omega_z}{dt} = \frac{d^2\theta}{dt^2} \tag{21}$$



(a) $\vec{\boldsymbol{\alpha}}$ and $\vec{\boldsymbol{\omega}}$ in the same direction: Rotation speeding up



(b) $\vec{\boldsymbol{\alpha}}$ and $\vec{\boldsymbol{\omega}}$ in the opposite direction: Rotation slowing down

Figure 8: Angular acceleration and angular velocity vectors of a rigid body

Linear Speed in Rigid-Body Rotation

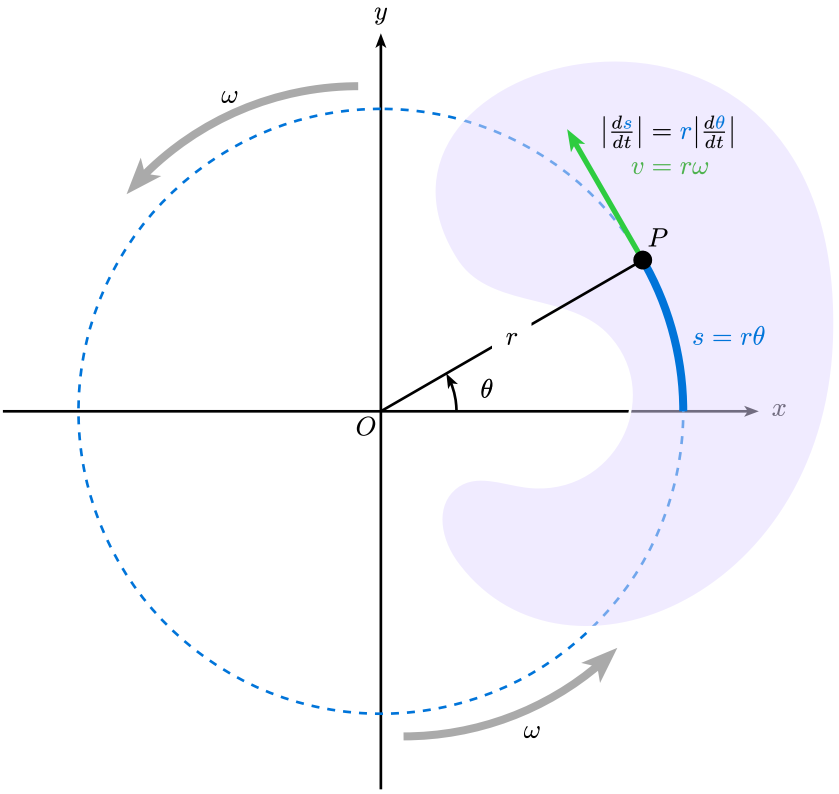


Figure 9: A rigid body rotating about a fixed axis through point O with linear speed v.

Linear Speed

"The farther a point is from the axis. the greater its linear speed."
(Young et al., 2016, Chapter 9.3, p.281)

$$v = r\omega \tag{22}$$

(Young et al., 2016, Chapter TBD)

Proof of Equation 22
At any time t, **Equation 14** relates the angle θ (in radians) and the arc length s:

$$s(t) = r \cdot \theta(t) \equiv m \cdot \text{rad} \tag{i}$$

We take the derivative of bothe sides with respect to time t, and take the absolute value of both sides:

$$\left|\frac{ds}{dt}\right| = r\left|\frac{d\theta}{dt}\right| \tag{ii}$$

Now, $\left|\frac{ds}{dt}\right|$ is the absolute value of the rate of change of the arc length i.e. equal to the instantaneous linear speed v of a particle.

$$\left|\frac{ds}{dt}\right| = v \equiv \frac{m \cdot \text{rad}}{s} \tag{iii}$$

Similarly, recalling **Equation 19**

$$\left|\frac{d\theta}{dt}\right| = \omega \equiv \frac{\text{rad}}{s} \tag{iv}$$

Thus, we have

$$v = r\omega \equiv m \cdot \frac{\text{rad}}{s} \tag{v}$$

■

Linear Acceleration
Acceleration $\vec{\boldsymbol{a}}$ of a particle moving in a circle can be decomposed into two components:

- Tangential acceleration $\vec{\boldsymbol{a}}_{\text{tan}}$
- Centripetal acceleration $\vec{\boldsymbol{a}}_{\text{rad}}$
- Note: **Equation 23** is representing scalar quantities, not vectors

Tangential Acceleration
- Tangent to the path

$$\begin{aligned} \frac{dv}{dt} &= r\frac{d\omega}{dt} \\ a_{\text{tan}} &= r\alpha \end{aligned} \tag{23}$$

Centripetal Acceleration
- Always points toward the axis of rotation

$$a_{\text{rad}} = \frac{v^2}{r} = \omega^2 r \tag{24}$$

Proof of Equation 24:

■

Periodic Motion

Oscillation: can only occur/always occurs when there is a **restoring force** ($\overrightarrow{\boldsymbol{F_r}}$) tending to return the sysem to equilibrium.

Key Words:

- Amplitude $\boldsymbol{A}$
- Cycle
- Period $\boldsymbol{T}$, time to complete one cycle
- Frequency $\boldsymbol{f}$, number of cycles per second $\frac{\text{cycle}}{s}$

$$f = \frac{1}{T} \equiv \text{Hz} = \frac{1 \text{ cycle}}{s} \tag{25}$$

- Angular Frequency $\boldsymbol{\omega}$

$$\omega = 2\pi f = \frac{2\pi}{T} \equiv \frac{\text{rad-cycle}}{s} \tag{26}$$

Simple Harmonic Motion (SHM)
Conditions:

- Displacement is sufficiently small $\vec{\boldsymbol{r}}(x, y, z)$ or $\vec{\boldsymbol{r}}(r, \theta)$ aka small angles
- Amplitude is sufficiently small
- Uniform circular motion

The simplest kind of oscillation occurs when the restoring force is directly proportional to the displacement ($\vec{\boldsymbol{r}}$) from equilibrium.

$$\overrightarrow{\boldsymbol{F_r}} = -k\vec{\boldsymbol{r}} \tag{27}$$

where $k > 0$ with units of N/m = kg /s^2 , assuming no friction.

Harmonic Oscillator: a system that undergoes SHM. Examples include a mass on a spring and a pendulum (**for small angles**).

Use cases - to approximate:

- vibration of tuning forks
- electric curent in AC circuits
- oscillations of atoms in molecules and solids

Hooke's Law:

- Conditions similar to SHM
- The restoring force is proportional to the displacement from equilibrium (Young et al., 2016, Chapter 6.3, p.184)
- The proportionality of stress and strain in elastic deformations (Young et al., 2016, Chapter 11.4-11.5)
- In elastic deformations, stress (force per unit area) is proportional to strain (fractional deformation), the proportionality constant is called the elastic modulus

$$\frac{\text{Stress}}{\text{Strain}} = \text{Elastic Modulus} \tag{28}$$

(Young et al., 2016, Chapter 14)

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